Elijah Wright Spiegel

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AOS

- Formal Epistemology
- Bayesian Game and Decision Theory
- Philosophy of Statistics

AOC

- Epistemology
- Logic
- Philosophy of Data and AI

Education

Ph.D. Logic and Philosophy of Science — University of California, Irvine 2023–Present

M.S. Statistics — University of California, Irvine 2026 (Expected)

M.A. Philosophy, Special Program in Symbolic Systems — Stanford University 2021

B.A. Philosophy — Stanford University 2020

Reviews

- Review of Gerhard Schurz's *Optimality Justifications*
  Philosophy of Science — Forthcoming

Under Review

(A † indicates a co-authored paper.)

- At the Edge of Putnam's Program: Limitative Results for Computable Inductive Logics (†)
  Micro-abstract: We find consistent Δ_2^0 probabilities defined over languages strong enough to express arithmetic and show this is the best one can do. This improves on existing approaches which are not even arithmetically definable.
- Expanding the Horizons of Meta-induction
  Micro-abstract: Schurz improved on Reichenbach's vindication of induction using meta-inductively optimal methods. Sterkenburg pointed out they can only predict one step into the future. I get them to predict further out.
- Verification and Merging (†)
  Micro-abstract: We defend the Vienna Circle's insight into the connection between verification and agreement without adopting verificationism by showing that coming to agreement on observation sentences entails coming to total agreement.

Drafts Available

- Theory-laden Evidence and Incommensurability in a Bayesian Setting
  Micro-abstract: I explicate the requisite concepts and, subject to the explications, prove the incommensurability of scientific theories in the theory-laden setting.

Invited Talks and Posters

- Theory-laden Evidence and Incommensurability in a Bayesian Setting (Poster) — Philosophy of Science Association 2026
  Micro-abstract: (See draft of the same title above)
- Social Choice with Noisy Preferences — Philosophy Politics and Economics 2026
  (Joint work with Christian Torsell)
  Micro-abstract: Self-reported preference rankings are noisy, so the voting method you think is ideal for true preferences can be different than the voting method you should apply to reported preferences.
- Bayesianism for the Indecisive (Poster) — Philosophy of Science Association 2024
  Micro-abstract: Wald's Complete Class Theorem supports an argument for Bayesianism that is folkloric among statisticians but not philosophically clear. I clarify the argument and prove a computable version of the theorem to make the argument stronger.
- Bayesianism for the Indecisive — U. of Minnesota Philosophy of Statistics Early Career Summer Workshop 2024
- Epistemology and the Foundations of Statistics — LPS Summer Diversity Program 2024
  Micro-abstract: Statistics without epistemology is uninterpreted mathematics. Epistemology, it turns out, is difficult.

Comments

- Comments on Patrick Cronin's "Misconceived Track Records about Unconceived Alternatives" — APA Central 2025

Teaching

Making Modern Science — UC Irvine
TA — Spr26

Critical Reasoning — UC Irvine
TA — Win25, Fall26

Naturalized Epistemology — UC Irvine
TA — Fall24

Introduction to Inductive Logic — UC Irvine
TA — Spr24, Spr25, Win26

Introduction to Symbolic Logic — UC Irvine
TA — Win24

Philosophy of Language — Stanford University
TA — Fall20

Service

Reviewer for Philosophy and Phenomenological Research

Awards

- Summer 2026 Graduate Student Research Award
- 2025-2026 Graduate Service Award
- Summer 2025 Graduate Student Research Award
- 2024-2025 Graduate Service Award

- Summer 2024 Graduate Student Research Award
- 2023-2024 Graduate Service Award